SOC Incident Response & Threat Hunting

NS-SOC-001 | Northstar Technologies | Simulated incident | 19 October 2026 (UTC)

Disposition: High priority (assumed lab severity) / suspected unauthorized account access and endpoint compromise / OPEN - requires investigation.

Provenance: Synthetic CSV observations are distinguished from supplemental Exercise 5 scenario assertions. No real attack, malicious destination, data exfiltration or completed containment has been verified.

Executive summary

On October 19, 2026, synthetic VPN gateway records show 18 unsuccessful authentications for account j.park from 203.0.113.77 between 13:00 and 13:08 UTC, followed by a successful VPN login at 13:10 UTC. At 13:14, endpoint WS-101 recorded PowerShell launched with an encoded command, and at 13:16 a child command executed whoami. The workstation also logged outbound TCP connections at 13:15 and 13:17 to 198.51.100.88:443. A separate classroom supplement states that EDR attributed those connections to PowerShell PID 4820 and that the employee denied initiating the activity. Together these facts justify urgent response, but account compromise, C2 and data loss remain unconfirmed.

Evidence and sequence

Original log observations: VPN failures (13:00-13:08), success (13:10); encoded PowerShell (13:14); DNS query to sync-check.northstar-lab.example with NOERROR (13:14); network connections to 198.51.100.88:443 (13:15, 13:17); child cmd.exe /c whoami (13:16). The DNS log does not contain answer records; the network log does not contain process IDs or payloads. Supplemental scenario only: EDR claims attribution to PID 4820; employee denies initiating activity; workstation has access to internal documents. See incident_timeline.csv for row-level references.

Assessment and uncertainty

High priority is reasonable under the exercise definition because successful authentication from the repeated-failure source is followed by suspicious endpoint execution. Encoded PowerShell and whoami are dual-use and do not alone confirm maliciousness. Observed bytes_out cannot establish exfiltration, and TCP port 443 cannot prove HTTPS or C2. The three documentation-range IP networks used in this dataset are fictitious identifiers, not external threat-intelligence indicators.

Recommended response: first 30 minutes

0-5 min: page IR lead, open incident bridge and assign incident owner; document triage source and evidence times. 5-10 min: preserve available VPN/MFA logs, EDR process tree and network flow records; capture relevant volatile artifacts if feasible and authorized. 10-20 min: under IR approval, coordinate targeted EDR isolation of WS-101 (using suitable forensic access) and IAM revocation of active VPN sessions plus credential/MFA protection as warranted. 20-30 min: scope other accounts/hosts, inspect account changes, file access and network telemetry; record owner, time, outcome and evidence for each action. These are recommended steps, not completed actions.

Investigation priorities

Acquire authenticated EDR process-to-network records with PID, host, process GUID and precise event times; retrieve full PowerShell command and associated process tree if available; obtain DNS answers or resolver caches and proxy/TLS metadata; confirm MFA success/failure, employee context, and privileged activity; determine whether document access or lateral movement occurred; collect corroborating evidence before any claim of C2 or exfiltration.

Validation and closure gates

Closure requires a documented disposition supported by evidence (confirmed benign or investigated incident), defined impact and affected assets/accounts, evidence preservation, verified account/session security, endpoint cleanup or rebuild when needed, validation scans/EDR health and no recurrence during a defined observation period, and incident owner approval. If

benign, document a credible explanation for the login/PowerShell/network chain, not merely an absence of further alerts. Unknown impact is not itself proof of safety.

MITRE ATT&CK mapping - analytical only

Possible mappings: PowerShell execution to T1059.001 (Command and Scripting Interpreter: PowerShell), and whoami to T1033 (System Owner/User Discovery). These classifications describe observed behaviors, not attribution to an adversary. The VPN authentication sequence could represent credential attacks but lacks enough evidence to assign a confident intrusion technique.

Limitations and reproducibility

Synthetic sources are in data/*.csv and soc_events.sqlite. Run python scripts/analyze_case.py; inspect evidence/incident_timeline.csv for CSV row numbers. Additional classroom evidence lives in evidence/scenario_supplements.csv and is not contained in the original logs. Sigma/SPL files are draft detection content and have not been production validated.